



eJPT - Certified Junior Penetration Tester

Letter Of Engagement

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Exam Lab Configuration

The ejPT Exam environment is an in-browser lab environment that provides you with access to a pre-configured Kali Linux system with all tools, scripts, and wordlists required to successfully identify and exploit vulnerabilities on hosts in the target network.

The in-browser lab architecture ensures that you can begin, progress, and complete the exam from any device, from any location with a stable internet connection, and without the need to set up your own Virtual Machine(s).

You will be provided with a pre-configured Kali Linux system accessible via RDP through your browser via Apache Guacamole.

The Kali Linux system provided to you and the target host(s) within the lab environment are not connected to the internet; as a result, you will be required to utilize your host operating system for research and note-taking.

All required exploitation and post-exploitation tools, scripts, and frameworks are accessible on the Kali Linux system. The Kali Linux system also provides you with access to exploit modules/code via the Metasploit Framework and the locally stored Exploit Database (Exploit-DB) accessible via **"SearchSploit"**.

NOTE: You do not need to download or install any custom scripts or tools onto the in-browser Kali Linux system you have been provided with access to; the Kali Linux system has everything you need to complete the exam successfully.

You can, however, copy raw exploit code into the Kali Linux system via your clipboard (For more information on this, please refer to the **"Lab Guidelines"** document.

Letter of Engagement

Client Overview

Syntex Dynamics is a small technology-focused organization that helps companies improve efficiency through the design and delivery of technology-driven solutions. The organization supports both internal business operations and externally facing services, enabling its clients to streamline processes, enhance scalability, and better integrate their systems.

To support its operations, Syntex Dynamics maintains a segmented network architecture. Public-facing assets, including web servers and externally accessible websites, are hosted within a Demilitarized Zone (DMZ).

Additionally, the organization operates a separate internal/private network that is segmented from the DMZ and is intended for development and staging activities. This internal environment is used to test applications, workflows, and supporting infrastructure before production deployment.

Assessment Objectives & Rationale

Syntex Dynamics is in the process of scaling its development operations to support new clients across multiple industries. As part of this growth, the organization must ensure that its technical environment meets the security requirements defined by its clients, as well as applicable regulatory and compliance obligations.

To support this objective, Syntex Dynamics has engaged in a penetration testing assessment to evaluate the security posture of its infrastructure. This assessment is intended to identify security weaknesses within the environment, assess the potential impact of exploitation, and provide actionable insight into areas requiring remediation or improvement.

This penetration test represents the organization's first formal security assessment of its environment. The results of this engagement will be used to establish a baseline understanding of the current security state, which will inform future security improvements and serve as a reference point for subsequent, recurring assessments.

Penetration Testing Objectives

The objective of this engagement is to assess the security posture of Syntex Dynamics' technical environment through a controlled penetration testing assessment. Your firm has been contracted to perform this assessment, and you have been assigned as the lead penetration tester for the engagement.

The primary goals of the penetration test are to identify potential security weaknesses and vulnerabilities within the scoped environment and to safely exploit these issues where possible. Exploitation activities will be conducted to demonstrate the real-world impact of a potential intrusion, including the risk to systems, data, and internal network segmentation.

Specifically, the objectives of this engagement are to:

- Assess the overall security of Syntex Dynamics' DMZ and internal network environment
- Identify misconfigurations, weaknesses, and exploitable vulnerabilities
- Demonstrate the potential impact of successful exploitation through controlled attacks
- Evaluate the effectiveness of network segmentation between the DMZ and internal networks

As part of the engagement setup, access will be provided to a jumpbox (Kali Linux system) located within the client's DMZ network, which will serve as the initial starting point for all testing activities. All assessment actions will originate from this system and remain within the agreed scope of the engagement.

Scope Of Engagement

You have been hired to perform an on-site Black-Box Penetration Test against the hosts, web applications, and networks of **Syntex Dynamics**.

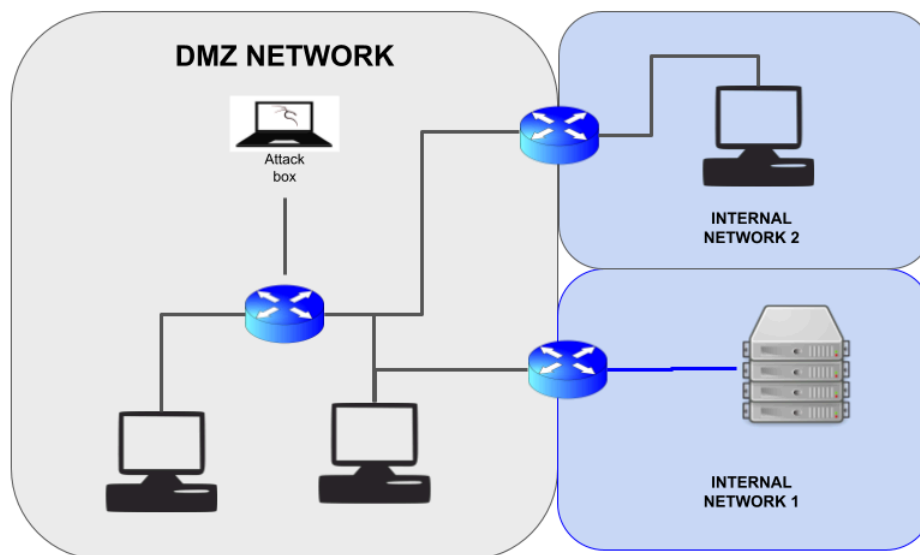
This is what the client organization has defined as the scope of the tests:

- The DMZ network you will be connected to.
- Any other internal network you will be able to reach/access.

When you first initiate your exam lab environment, you will automatically be connected to the DMZ network. This network contains production servers and web applications that you can potentially gain access to via brute-force attacks and via the exploitation of misconfigurations or vulnerabilities.

You are also required to identify any internal networks, pivot into them, identify hosts within these internal network/networks, and gain access to these systems.

Please note that the following diagram is a guideline and does not reflect the actual exam network configuration.



From the DMZ network, it is possible to reach some other internal networks/network. All the networks are /24 networks. You have to find the routing information and the network addresses by identifying a pivot point that interconnects the two networks.

The Kali Linux system contains all the dictionaries of common usernames and passwords, should you have to perform a brute force attack.

Exam Objectives

Please note that you are required to perform a penetration test on all the hosts in scope.

The questions in the test area will cover most of your eJPT findings. You can freely choose how and in what order you want to answer the questions during the penetration test.

Keep in mind that both the quiz area and the exam lab environment will be accessible simultaneously for the entire period of the exam (48 Hours).

Recommended Tools

You are free to use any environment to perform your penetration test.

The following is a suggested list of tools that might be useful during the exam:

- Nmap
- Dirbuster
- Nikto
- WPScan
- CrackMapExec
- The Metasploit Framework
- Searchsploit
- Hydra

Please note that this list of tools is simply a list of recommendations; this list does not serve as explicit instructions or hints on what tools to use in any way. Feel free to use any tool you are comfortable with to facilitate your penetration test.

Furthermore, all tools mentioned above, and a few select others, have been pre-installed on the Kali Linux system provided to you.

Recommended Wordlists

The exam lab environment may contain a few web applications, servers or services that can be attacked via a brute-force/password-spraying attack.

As a result, it is imperative that you ensure you maximize the efficiency of your brute-force/password-spraying attacks.

Be very careful with your time. Any host/server that can be accessed successfully via a brute-force/dictionary attack will only take a couple of minutes to complete. If a brute-force/dictionary attack takes more than 20-30 minutes to complete, you are likely using the incorrect wordlist(s).

In the interest of efficiency of your assessment, the following is a list of recommended wordlists to use during your engagement:

Passwords:

- `/usr/share/wordlists/rockyou.txt`

Hints For A Successful Penetration Test

- Do not think about the exam as a "Capture the Flag." It is not.
- Refer to the **Lab Guidelines** document provided to you along with this document for information on how the in-browser lab works.
- Take notes and save your findings on an external note-taking application on your host operating system. (This is because any data you store on the Kali Linux system will be deleted if you reset your lab environment.)
- Another great idea is to answer the questions while you proceed with your Penetration Test. (Carefully read the question, then perform your tests in the lab)
- Any attack, tool, or technique, according to the rules of engagement, is allowed during the exam.